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# Week 4, Lecture 3
# Quantum communication

References:
Nielsen & Chuang ch. 1.3, 12.6
Bachor & Ralph ch. 13

# What we will cover today…

01

**Quantum teleportation**

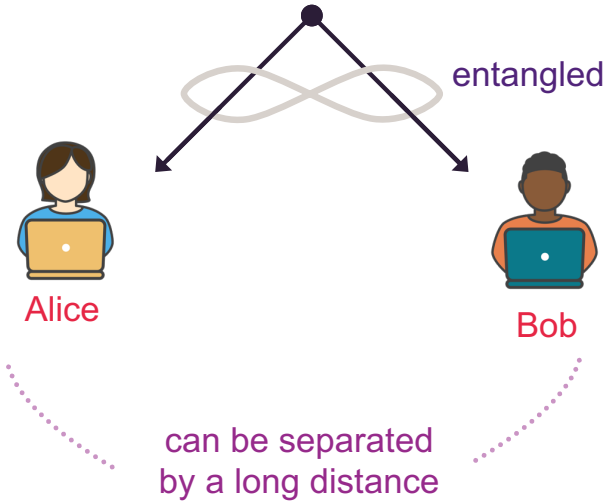
02

**Quantum dense coding**

03

**Quantum key distribution**

# Entanglement as a resource for quantum communication




If Alice and Bob share an entangled pair, they can perfom

- Quantum teleportation
- Quantum key distribution
- Dense coding

…and other quantum information protocols

- In both quantum teleportation and dense coding, an entangled resource is required to be distributed between two spatially separated parties
- In fact, there are many quantum information protocols that also make use of distributed entanglement
- For a quantum channel between two parties to be useful for quantum communication, it must be able to preserve entanglement

→ the ability to distribute entanglement is necessary for quantum communication

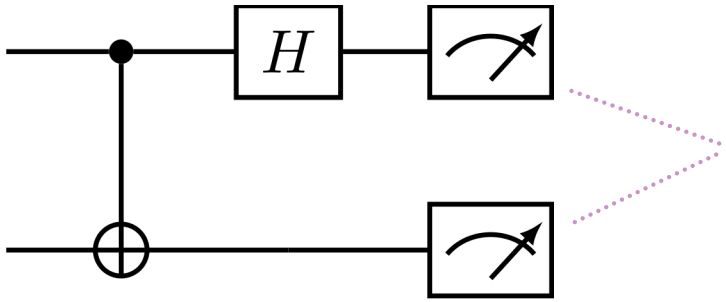
# Bell states and Bell state measurements - review

**Bell states**

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

maximally entangled two qubit states

**Bell state measurement**




measure in the computational basis

Input the four bell states into this circuit

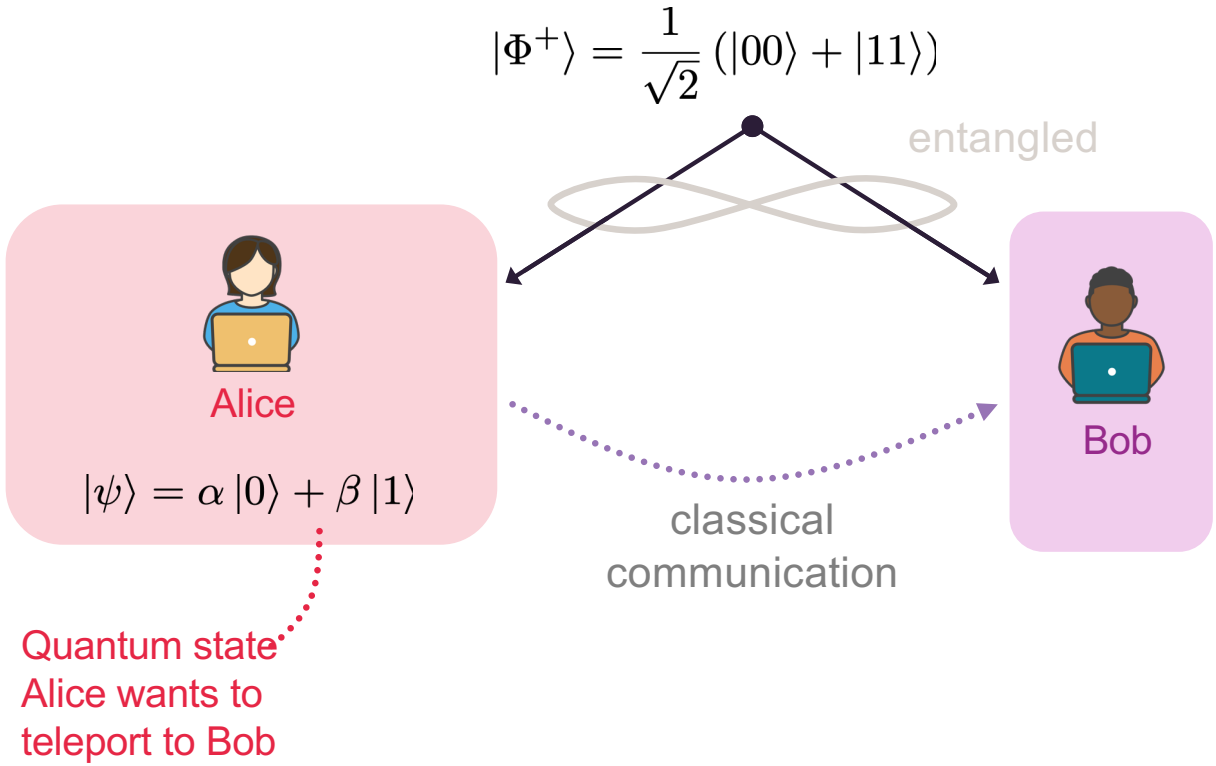
The computational basis measurement results at the end will be:

$$|\Phi^+\rangle \dashrightarrow |00\rangle$$
$$|\Psi^+\rangle \dashrightarrow |01\rangle$$
$$|\Phi^-\rangle \dashrightarrow |10\rangle$$
$$|\Psi^-\rangle \dashrightarrow |11\rangle$$

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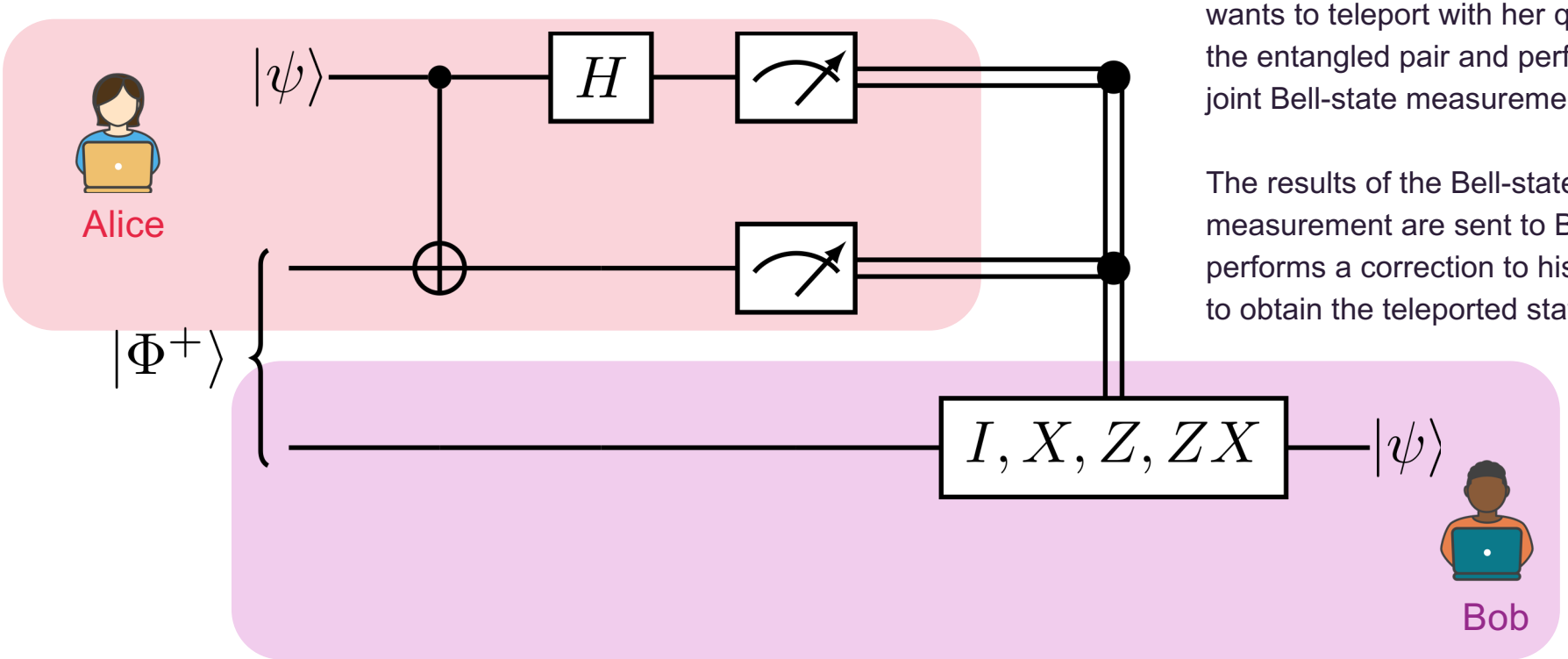
# Quantum teleportation

# Quantum teleportation




*Quantum teleportation*: by sharing an entangled resource state $|\Phi^+\rangle$ and a classical communication channel, Alice can transfer the state of a qubit to Bob.

# Quantum teleportation




Alice and Bob share an entangled pair $|\Phi^+\rangle$.

Alice combines the state that she wants to teleport with her qubit of the entangled pair and performs a joint Bell-state measurement.

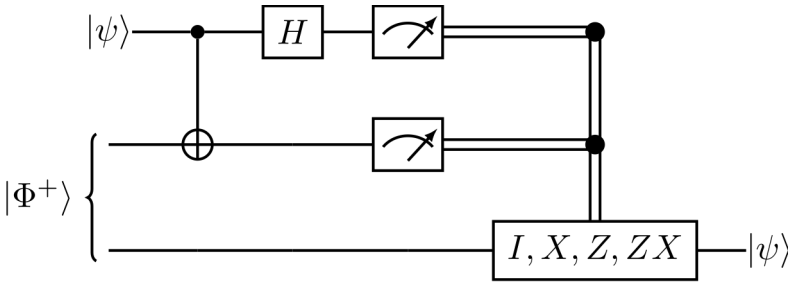
The results of the Bell-state measurement are sent to Bob, who performs a correction to his qubit to obtain the teleported state.

# Quantum teleportation circuit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \qquad |\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$|\psi\rangle \otimes |\Phi^+\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$= \frac{1}{\sqrt{2}}[\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)]$$




CNOT between first two qubits:

$$\rightarrow \frac{1}{\sqrt{2}}[\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|10\rangle + |01\rangle)]$$

Hadamard on first qubit:

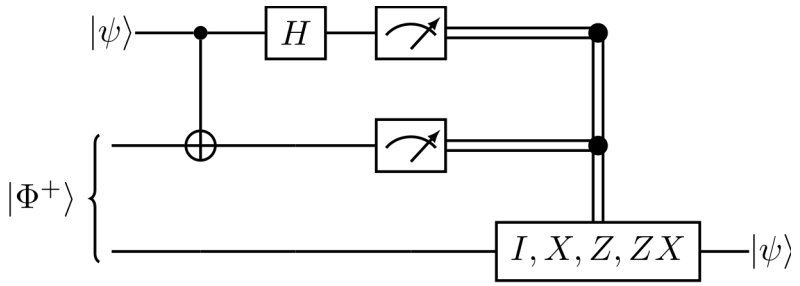
$$\rightarrow \frac{1}{2}[\alpha(|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta(|0\rangle - |1\rangle)(|10\rangle + |01\rangle)]$$

This state can be rewritten:

$$\frac{1}{2}[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)]$$

# Quantum teleportation circuit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \qquad |\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$




After the Hadamard gate:

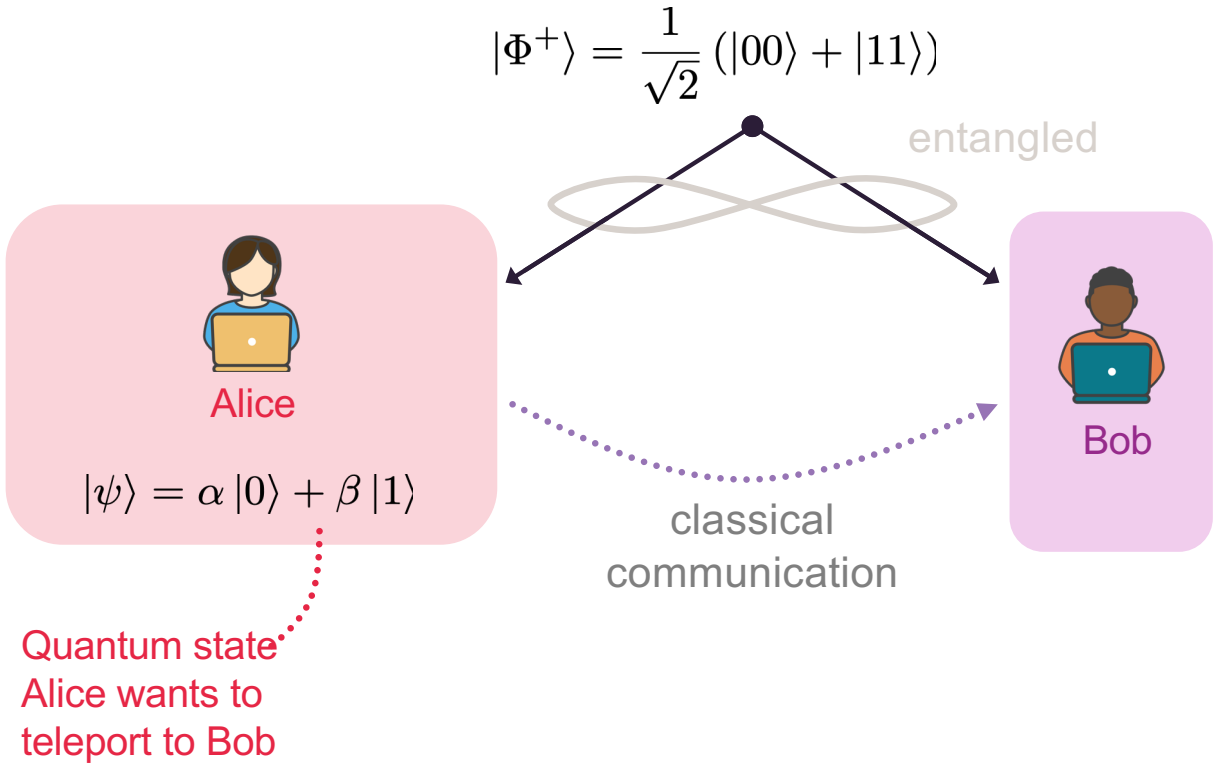
$$\frac{1}{2}\left[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)\right]$$

| measurement outcome | state of third qubit | correction |
|---|---|---|
| $\|00\rangle$ | $\alpha\|0\rangle + \beta\|1\rangle$ | $I$ |
| $\|01\rangle$ | $\alpha\|1\rangle + \beta\|0\rangle$ | $X$ |
| $\|10\rangle$ | $\alpha\|0\rangle - \beta\|1\rangle$ | $Z$ |
| $\|11\rangle$ | $\alpha\|1\rangle - \beta\|0\rangle$ | $ZX$ |

The first two qubits are measured in the computational basis.

From the measurement results, we know the state of the third qubit and can perform the required correction $\{I, X, Z, ZX\}$ to recover the correct qubit state.

# Quantum teleportation - observations




Is the *no-cloning theorem* violated?
No, because Alice's qubit is destroyed in the measurement process. Her qubit's state is transferred to Bob, not copied.

Is Alice's qubit teleported instantly to Bob?
No: Alice needs to send Bob the (classical) information on her measurement results before he can use them.

Is this protocol useful at all?
Yes, because Alice can "send" Bob a qubit by just transmitting 2 bits of classical information.

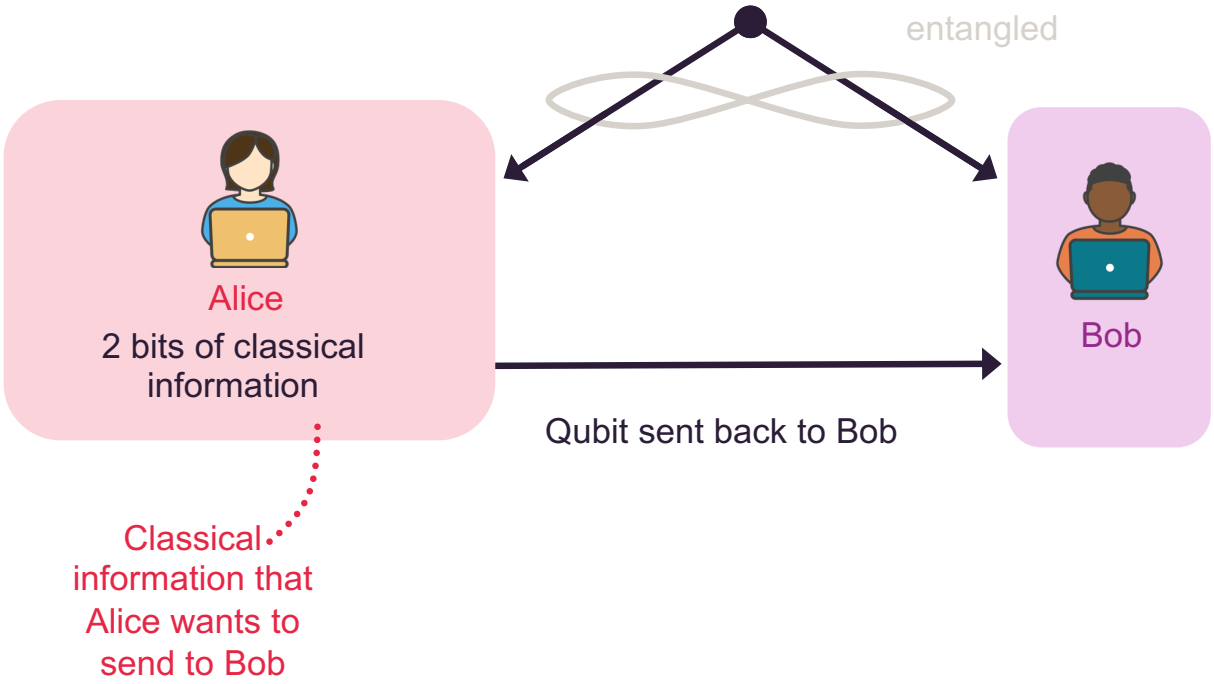
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# Quantum dense coding

# Quantum dense coding

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

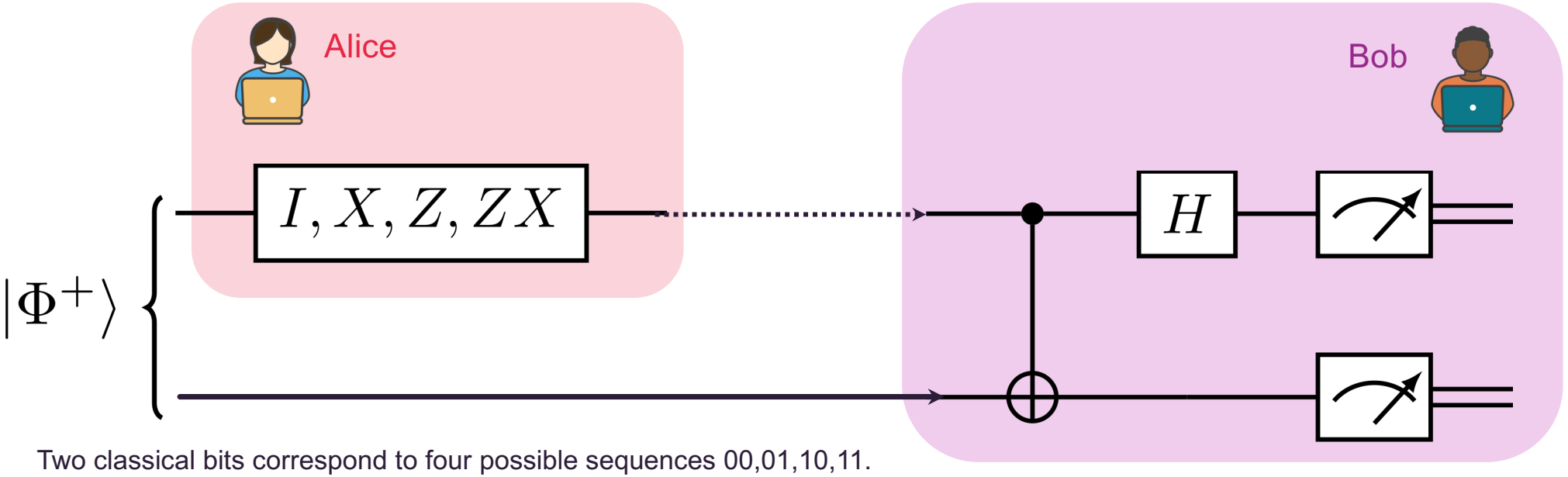



Quantum dense coding can be thought of as the reverse protocol of quantum teleportation.

In quantum teleportation, one qubit is sent from Alice to Bob by making use of an entangled pair, and sending 2 classical bits of information.

In quantum dense coding, 2 classical bits of information are sent from Alice to Bob using an entangled pair, where Alice performs some operation to her qubit of the entangled pair and sends the qubit back to Bob.

# Quantum dense coding

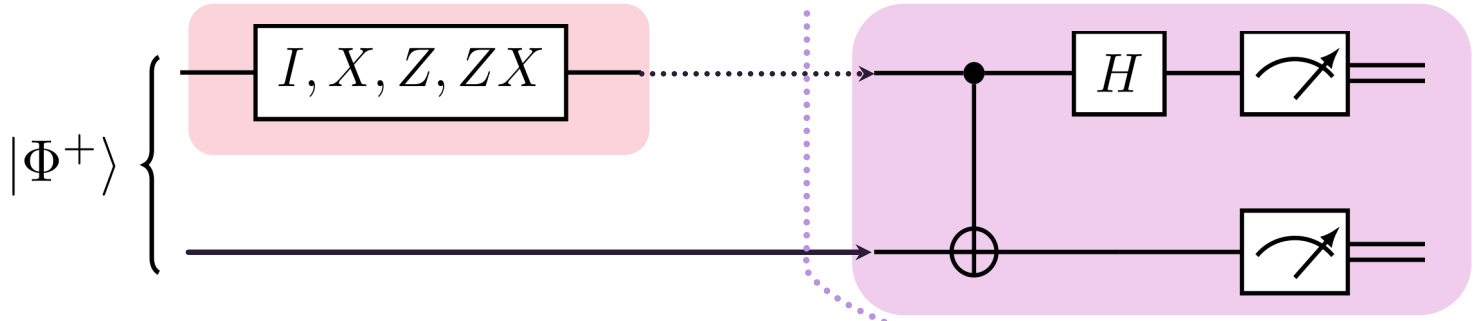



Two classical bits correspond to four possible sequences 00,01,10,11.

The two bits of classical information Alice wants to send to Bob, determines which of the operations $\{I, X, Z, ZX\}$ she performs on her qubit of the entangled pair.

Her qubit is then sent to Bob who performs a joint Bell state measurement to work out the two bits of classical information Alice has sent.

# Quantum dense coding



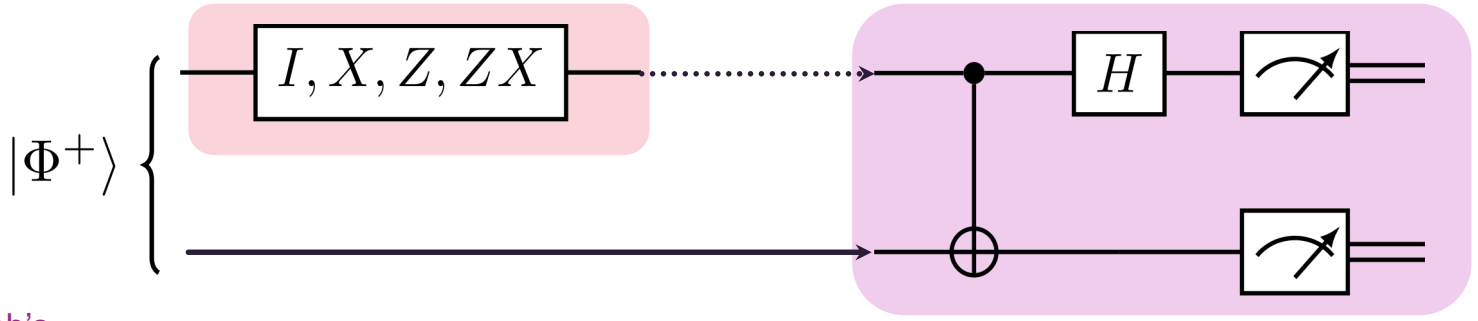

| classical bit strings | Alice's operation | state of bob's entangled pair |
|---|---|---|
| 00 | $I_A \|\Phi^+\rangle = \frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle) = \|\Phi^+\rangle$ | |
| 01 | $X_A \|\Phi^+\rangle = \frac{1}{\sqrt{2}}(\|10\rangle + \|01\rangle) = \|\Psi^+\rangle$ | |
| 10 | $Z_A \|\Phi^+\rangle = \frac{1}{\sqrt{2}}(\|00\rangle - \|11\rangle) = \|\Phi^-\rangle$ | |
| 11 | $Z_A X_A \|\Phi^+\rangle = \frac{1}{\sqrt{2}}(\|10\rangle - \|01\rangle) = \|\Phi^-\rangle$ | |

Shown on the left are the four possible bit strings of the two bits of classical information that Alice wants to send to Bob.

She then performs a Pauli operation to her qubit of the entangled pair and sends that qubit to Bob.

The possible states of Bob's two entangled qubits are shown here.

# Quantum dense coding




state of bob's entangled pair | CNOT | Hadamard | classical bit strings

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \rightarrow \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \rightarrow |00\rangle \dashrightarrow 00$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle) \rightarrow \frac{1}{\sqrt{2}}(|11\rangle + |01\rangle) \rightarrow |01\rangle \dashrightarrow 01$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \rightarrow \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) \rightarrow |10\rangle \dashrightarrow 10$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|10\rangle - |01\rangle) \rightarrow \frac{1}{\sqrt{2}}(|11\rangle - |01\rangle) \rightarrow |11\rangle \dashrightarrow 11$$

Bob now peforms a Bell state measurement on both the qubits to discover the classical two-bit string that Alice sent.

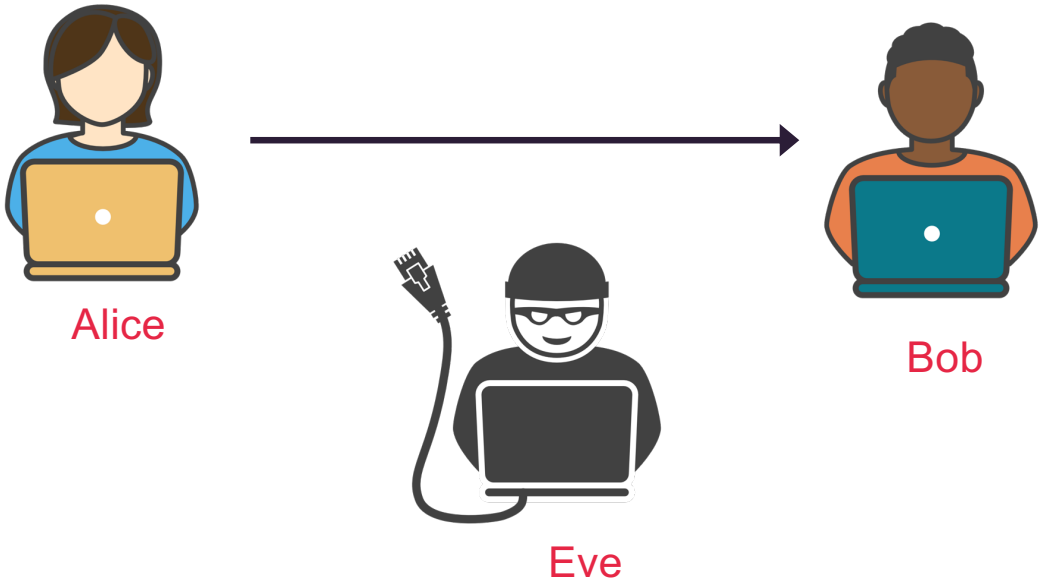
The measurement outcome of both qubits in the computational basis corresponds to the two bits of classical information.

How can Alice send 2 bits of information with only one qubit?
Alice and Bob were sharing an entangled state in the beginning; the 2 bits are encoded in the bipartite state.

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# Quantum key distribution

# Using quantum mechanics for secure communication

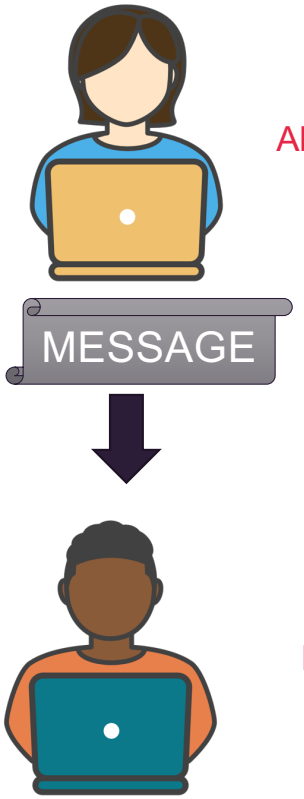



Alice wants to send a secret message to Bob.

An eavesdropper (Eve) wants to intercept that message.

Quantum mechanics can be used for cryptography to enable Alice and Bob to communicate securely.

# One time pad (Vernam cypher)




Alice

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| original message | h | e | l | l | o | | | |
| message converted to ASCII binary | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 … |
| | ⊕ | ⊕ | ⊕ | ⊕ | ⊕ | ⊕ | ⊕ | ⊕ |
| private key | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 1 … |
| encrypted message | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 … |

send to Bob

Bob

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| encrypted message | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 … |
| | ⊕ | ⊕ | ⊕ | ⊕ | ⊕ | ⊕ | ⊕ | ⊕ |
| private key | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 1 … |
| decoded message | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 … |
| message converted to string | h | e | l | l | o | | | |

A *one time pad* is a type of private key cryptography scheme in which a message is encrypted using a secret key.

$\oplus$ is the bitwise XOR (exclusive or):

| $A$ | $B$ | $A \oplus B$ |
|---|---|---|
| 1 | 1 | 0 |
| 1 | 0 | 1 |
| 0 | 1 | 1 |
| 0 | 0 | 0 |

If Eve intercepts the message, she will not be able to understand it without the key.

# One time pad (Vernam cypher)

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The one-time pad is secure as longs as:

1. The secret key is at least as long as the plain text
2. Key must be random
3. Key can never be reused
4. Key must be completely secret

So how can we use quantum mechanics to distribute a completely secure secret key?

# Quantum key distribution

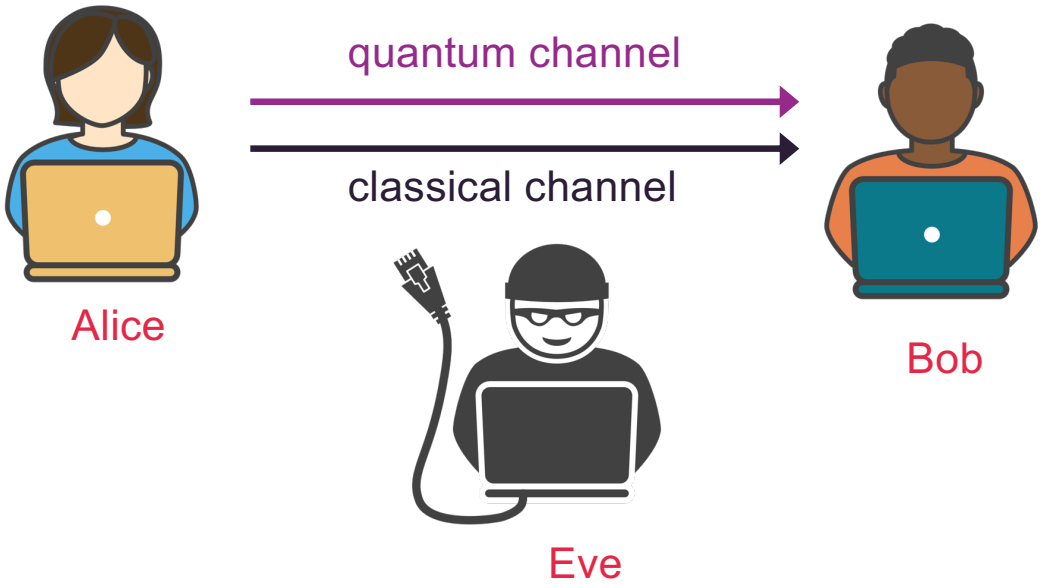



Quantum key distribution is a protocol that lets Alice and Bob share a secret key that can be used for cryptography.

In this process, the key is encoded in quantum states (photonic qubits).

The security of this protocol is based on the laws of quantum mechanics: any measurement an eavesdropper (Eve) makes on the quantum system changes the state of the system, therefore, Alice and Bob notice Eve's presence.

# BB84 protocol




Two bases $A = \{|a_0\rangle, \dots, |a_{d-1}\rangle\}$ and $B = \{|b_0\rangle, \dots, |b_{d-1}\rangle\}$ are mutually unbiased if $\left|\langle a_i|b_j\rangle\right|^2 = \frac{1}{d}\ \forall i, j \in 0, \dots, d-1$ .

Example: in $d = 2$, $\{|0\rangle, |1\rangle\}$ and $\{|+\rangle, |-\rangle\}$ are MUBs.

BB84 (by Charles H. Bennett and Gilles Brassard in 1984) is a "*prepare-and-measure*" QKD protocol.

The protocol uses two mutually unbiased bases (MUBs) which contain two states each, so four possible states in total.

Alice prepares a random quantum state among these four and sends it to Bob, who measures it in one of the two bases, chosen randomly.

Then they communicate through a classical channel to extract the key from the measurement outcomes.

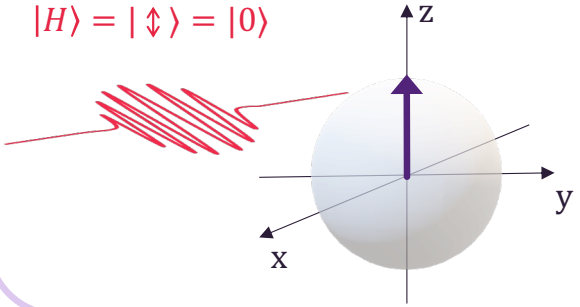
Eve could eavesdrop both on the quantum channel and on the classical channel.

# Photon polarisation - states

$\{|0\rangle, |1\rangle\}$ basis

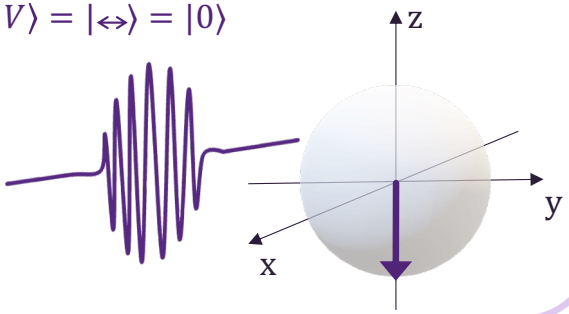
Horizontally polarised

$|H\rangle = |\updownarrow\rangle = |0\rangle$




Vertically polarized

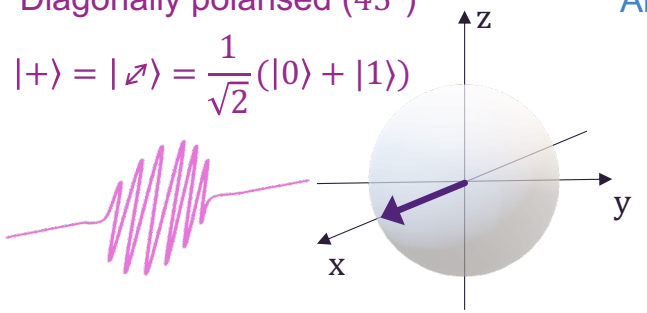
$|V\rangle = |\leftrightarrow\rangle = |0\rangle$




$\{|+\rangle, |-\rangle\}$ basis

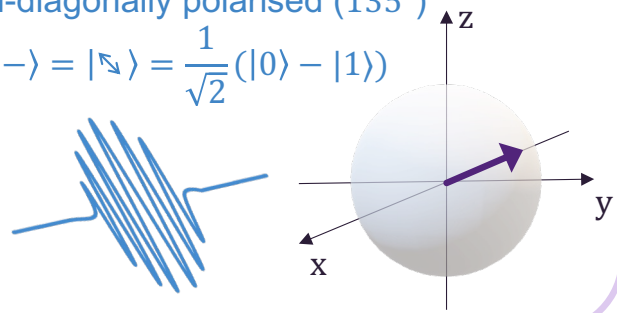
Diagonally polarised $(45°)$

$|+\rangle = |\nearrow\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

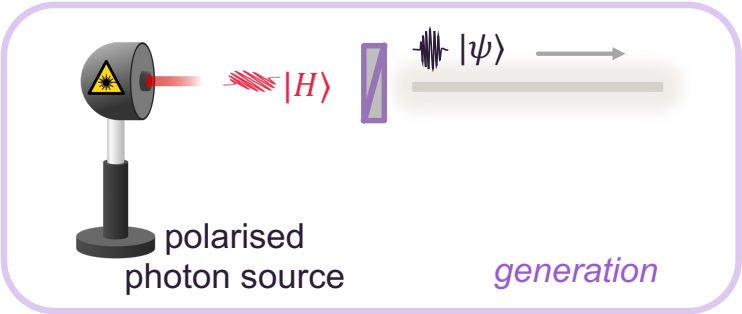



Anti-diagonally polarised $(135°)$

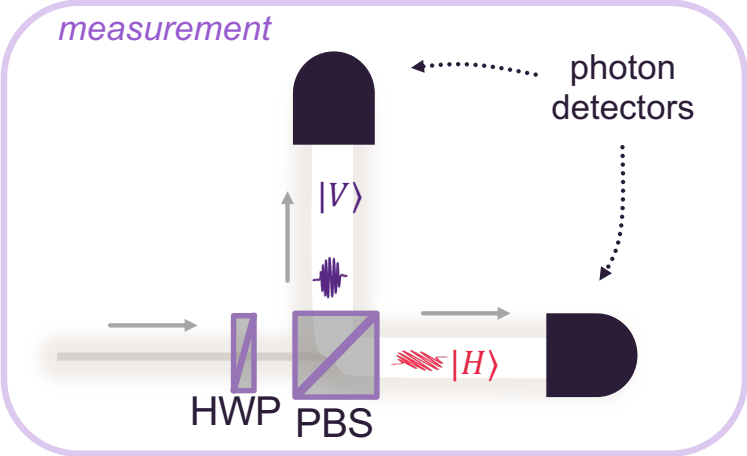
$|-\rangle = |\nwarrow\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$



# Photon polarisation – generation and measurement




Arbitrary linearly polarised states can be generated using a polarised photon source and a HWP at an arbitrary angle.

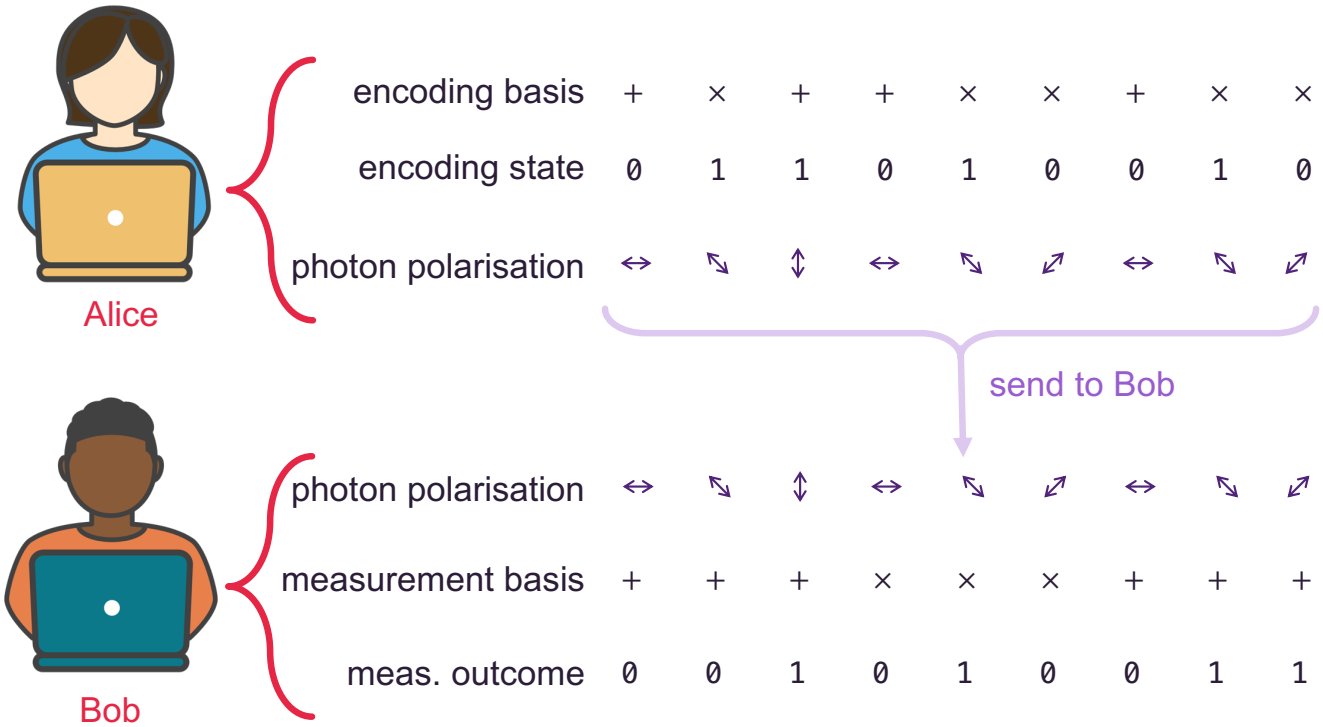



To measure polarisation qubits, a polarising beam-splitter splits light into two different modes each containing orthogonal polarisation states.

Photon detectors at the end of each output of the polarising beam splitter are needed to perform measurement of the qubit in the computational basis.

A HWP can rotate the measurement basis to the diagonal one.

# BB84 – 1. Raw key exchange



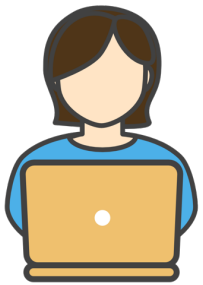

This is the only step that uses the quantum channel!

In each round:

1. Alice randomly chooses a basis.
2. Alice randomly chooses a state.
3. Alice encodes the chosen qubit and sends it to Bob.
4. Bob randomly chooses a basis.
5. Bob measures the incoming qubit in the selected basis.

After this step, Alice's encoding states and Bob's measurement outcomes form the *raw key*.

# BB84 – 2. Sifting



Alice

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| encoding basis | + | × | + | + | × | × | + | × | × |
| encoding state | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 |
| photon polarisation | ↔ | ⤡ | ↕ | ↔ | ⤡ | ⤢ | ↔ | ⤡ | ⤢ |
| Alice's sifted key | 0 | | 1 | | 1 | 0 | 0 | | |



Bob

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| photon polarisation | ↔ | ⤡ | ↕ | ↔ | ⤡ | ⤢ | ↔ | ⤡ | ⤢ |
| measurement basis | + | + | + | × | × | × | + | + | + |
| meas. outcome | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 |
| Bob's sifted key | 0 | | 1 | | 1 | 0 | 0 | | |

Alice's and Bob's raw keys do not match: when Bob chose a different basis than Alice, he obtained a random measurement outcome, which generated an error in 50% of cases.

In the *sifting* step, Alice and Bob communicate using the classical channel and compare their basis choices.

They eliminate all measurements in which they chose different bases, and obtain the *sifted key*. Ideally, the sifted key should match between Alice and Bob.

# BB84 – Intercept and resend attack




| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| encoding basis | + | × | + | + | × | × | + | × | × |
| encoding state | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 |
| photon polarisation | ↔ | ⤡ | ↕ | ↔ | ⤡ | ⤢ | ↔ | ⤡ | ⤢ |
| photon polaris. | ↔ | ⤡ | ↕ | ↔ | ⤡ | ⤢ | ↔ | ⤡ | ⤢ |
| meas. basis | + | × | × | × | × | + | + | + | × |
| meas. outcome | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| photon polaris. | ↔ | ⤡ | ⤡ | ⤡ | ⤡ | ↕ | ↔ | ↔ | ⤢ |
| photon polaris. | ↔ | ⤡ | ⤡ | ⤡ | ⤡ | ↕ | ↔ | ↔ | ⤢ |
| measurement basis | + | × | + | + | × | × | + | × | × |
| meas. outcome | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 1 | 0 |

But what if Eve tries to intercept the key?

Let us consider a type of attack called “intercept and resend”: Eve measures the photons in a random basis and then prepares other photons in the state she measured and sends them to Bob.

If she measures in the same basis that Alice prepares, then the state remains the same.

If she measures in a different basis, her outcome is random and she will send a different state to Bob. This can lead to errors in the sifted key.

(in the example, we only look at cases where Alice’s and Bob’s bases match for simplicity)

# BB84 – 3. Key distillation



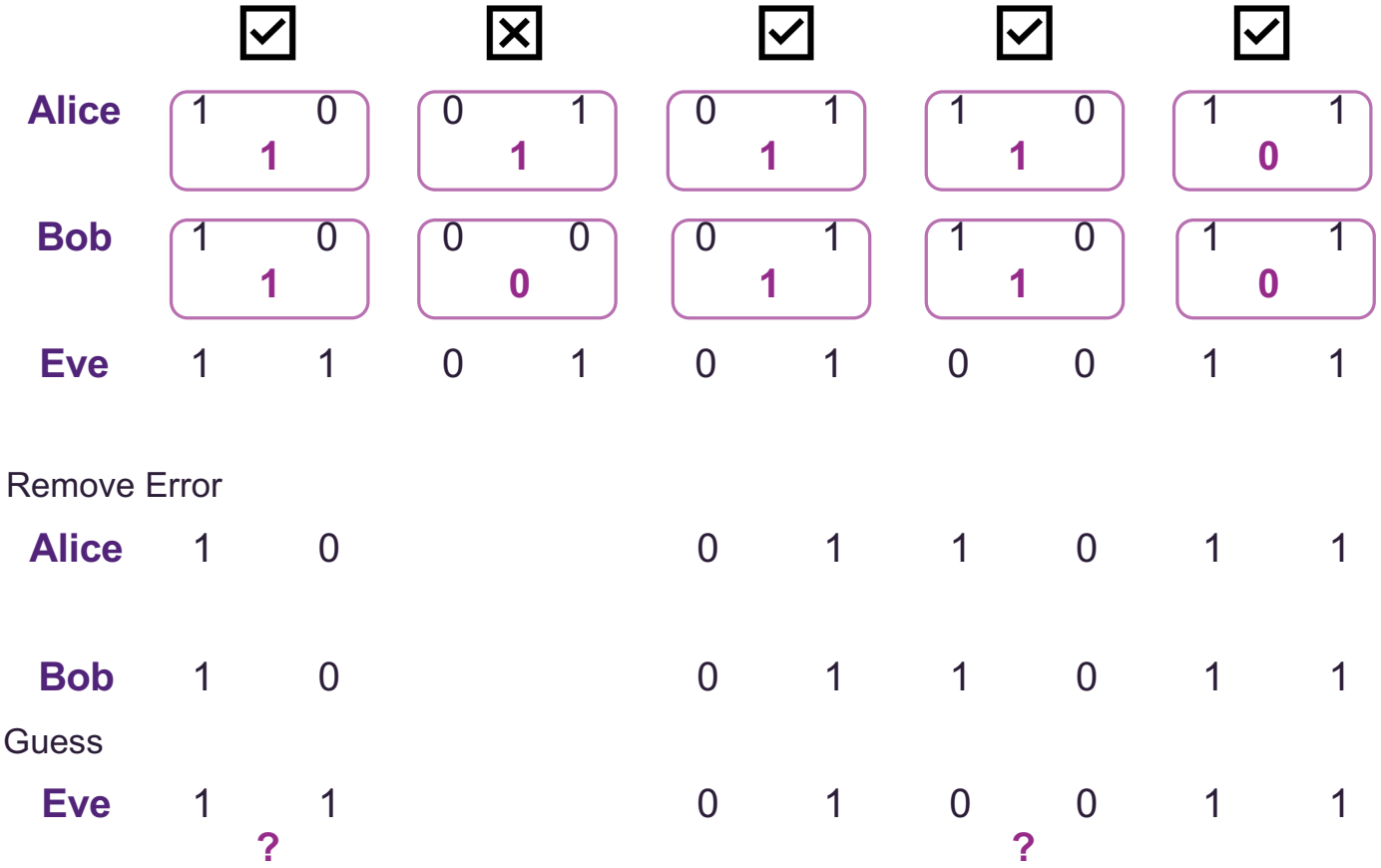

In the key distillation step (also known as *error correction*) Alice and Bob compare a subset of their bits from the sifted key to estimate the fraction of errors, called *quantum bit error rate* (QBER).

If they see errors, this signals the presence of an eavesdropper.

An intercept-resend attack would introduce a QBER of 25%. However, Eve could have better technology and perform different attacks that introduce less error. Alice and Bob never know the causes of errors; therefore, **they must attribute all errors to Eve** (even if they are caused by environmental noise).

A secure key can be extracted even in the presence of error! If Alice and Bob estimate that they share more information with each other than Eve could have obtained, they proceed with the rest of the protocol. First, they correct their keys to remove errors using classical statistics methods.

# QKD – Hashing




Hashing is an example of method to correct for errors between Alice's and Bob's keys without disclosing too much information about the secret key.

For example, by announcing the parity of pairs of bits, Alice and Bob can find and correct possible mistakes.

Different: Odd parity = 1
Same: Even parity = 0

In the end there will be no errors between Alice and Bob's bit strings.

Every bit of information shared here must eventually be removed from the key, which gets shorter!

# BB84 – 4. Privacy amplification

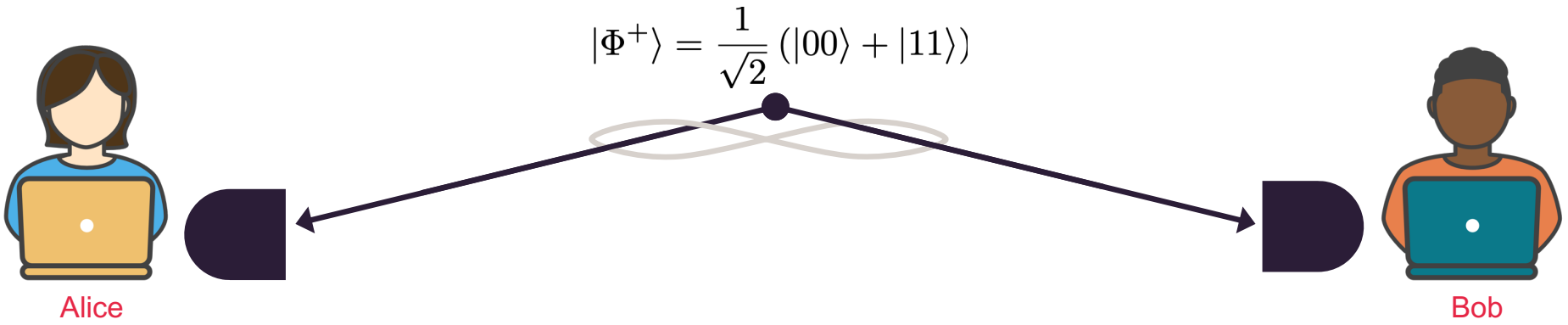
After the key distillation step, Alice's and Bob's keys are identical.

Eve still has some information on the key.

At this point, Alice and Bob perform *privacy amplification*: they use classical algorithms to decrease Eve's information on the key, at the cost of discarding key bits. Hashing could be used for privacy amplification as well. They do this until they are reasonably sure that **Eve does not have any information** on the key at all.

The key they share after privacy amplification is the *secret key*.

# Quantum key distribution – entanglement based




- Alice and Bob can also share a secret key using an ‘entanglement based’ QKD scheme.
- Many pairs of entangled qubits are distributed between Alice and Bob.
- They agree to measure a subset of the qubits in prearranged bases, either $\{|0\rangle, |1\rangle\}$ or $\{|+\rangle, |-\rangle\}$, chosen at random and compare results.
- If the measurement outcomes are perfectly correlated, then they know the channel is secure and there is no eavesdropper.
- This is equivalent to the ideal prepare-and-measure BB84 scheme.

# QKD – Absolute security

For the protocol to produce secret key, we require

$$K = \beta I(a:b) - I(E:A) > 0$$

$K$ Average number of distillable secret bits per use of the channel (in the asymptotic limit)

$\beta$ is the efficiency with which they implement privacy amplification

$I(a:b)$ is the classical mutual information between Alice and Bob's bit strings

$$I(a:b) = \sigma_c(a) + \sigma_c(b) - \sigma_c(a,b)$$

$I(E:A)$ is the quantum mutual information between Alice's measurement outcomes and Eve's state

$$I(E:A) = \sigma_q(E) - \sigma_q(E|a)$$

**What does $I(E:A)$ mean and how do we determine it?**

# Interpretation of quantum mutual information

**Alice** $\rho_{AE}$ **Eve**

Reduced state of Eve's mode is $\rho_E = \mathrm{Tr}_A(\rho_{AE})$

Alice gets measurement result $|a\rangle$

Eve's state conditioned on Alice's measurement result is $\rho_{E|a}$

Max information that Eve can extract about a's by doing optimal measurement on $\rho_{E|a}$ is

$$\begin{aligned} I(E:A) &= \sigma_q(\rho_E) - \sum P_a \sigma_q(\rho_{E|a}) \\ &= \sigma_q(\rho_E) - \sigma_q^a(\rho_{E|a}) \end{aligned}$$

If Eve was not correlated with Alice at all, there would be no difference between $\rho_E = \mathrm{Tr}_A(\rho_{AE})$ and $\rho_{E|a}$ :

$$\sigma_q(\rho_E) - \sum_a P_a \sigma_q(\rho_{E|a}) = 0$$

# Determining Eve's information

Assume $\rho_{ABE}$ is pure

$$\sigma_q(E) = \sigma_q(A,B)$$

$$\sigma_q(E|a) = \sigma_q(B|a)$$

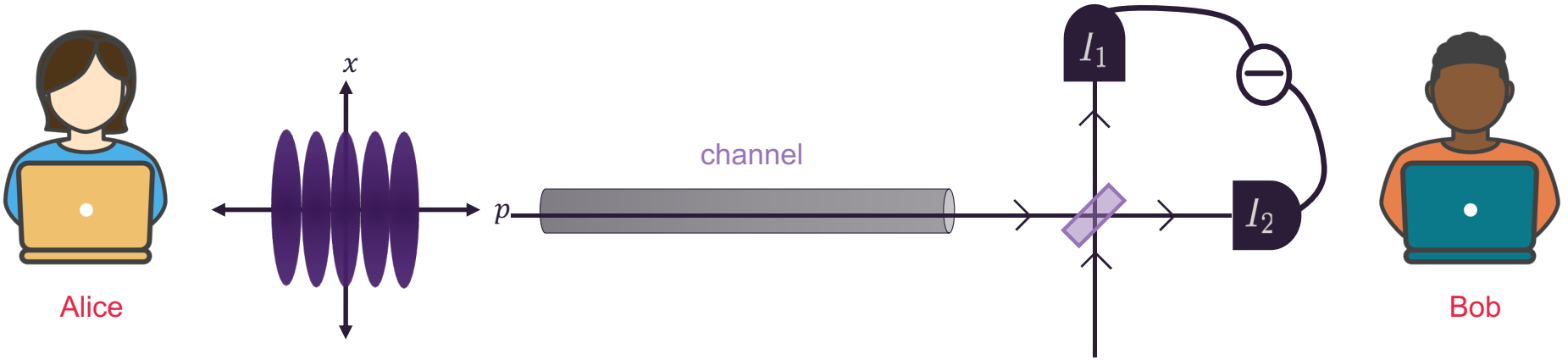
therefore

$$\begin{aligned} I(E:A) &= \sigma_q(E) - \sigma_q(E|a) \\ &= \sigma_q(A,B) - \sigma_q(B|a) \end{aligned}$$

$$K = \beta I(a:b) - I(E:A) > 0$$

# Continuous variable quantum key distribution

## Prepare and measure scheme



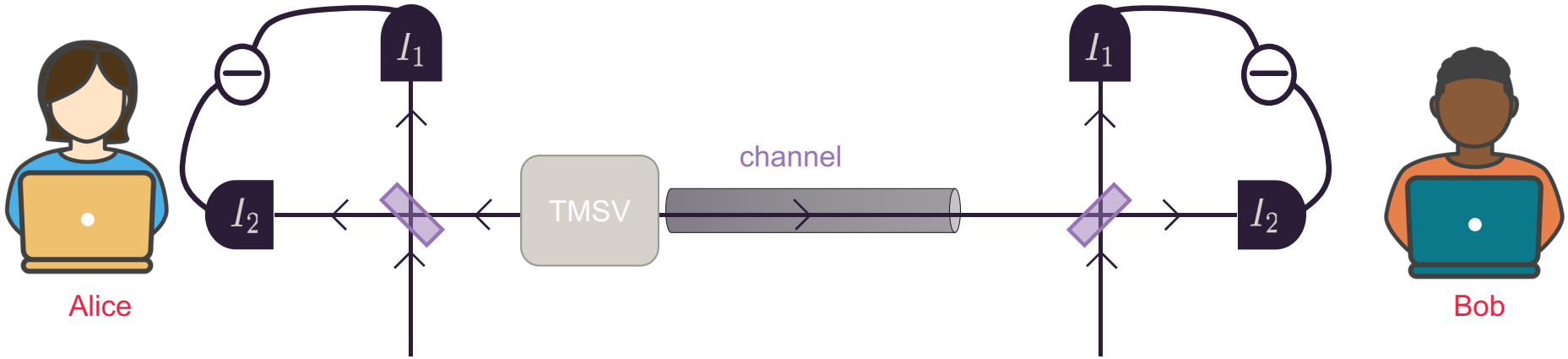

In a prepare and measure scheme for CVQKD, Alice prepares squeezed states modulated in randomly selected quadarature ($\hat{x}$ or $\hat{p}$) and sends them to Bob.

Bob then performs homodyne detection measuring a randomly selected quadrature, either $\hat{x}$ or $\hat{p}$.

They compare quadratures over a classical communication channel and keep instances where the preparation and measurement quadratures are the same.

# Continuous variable quantum key distribution

## Entanglement based




In an entanglement-based scheme for CVQKD, Alice prepares a TMSV entangled state and sends one of the entangled modes through the channel to Bob.

They both perform homodyne detection on their entangled modes.

This scheme is equivalent to the previous prepare and measure scheme where Alice prepares and sends squeezed states.